

# Conserved Pan-Fibrotic Core Transcriptomic Program & Multi-Model Consensus ML Biomarkers

Universal Extracellular Matrix Architecture Across Human Kidney, Liver, Lung, and Skin Biopsies (N = 1,069)

## A. Cross-Organ Discovery & Core Filtering

### 14 Human Discovery Cohorts (N = 1,069 Biopsies)

Kidney (4) | Liver (3) | Lung (4) | Skin (3)  
Moderated eBayes |log2FC| >= 0.585, FDR p < 0.05



### Conserved 4-Organ Core: 86 DEGs

Simultaneous statistical significance across all 4 anatomically distinct organ systems



### Matrisome Clean Core: 24 ECM Genes

Collagens, Regulators, Secreted Factors, Glycoproteins  
(Human Matrisome Masterlist Overlap)

## B. Pre-ML Multi-Center Replication

### Validation 1: Internal Replication (N = 4, n = 384)

- 24 / 24 Genes (100%) Concordant Direction in All 4 Organs
- GSE200818 (Kidney) · GSE162694 (Liver)
- GSE24206 (Lung) · GSE58095 (Skin)



### Validation 2: Held-Out Replication (N = 4, n = 105)

- 18 / 24 Genes Replicated in >= 2/4 Organs (Mann-Whitney U)
- GSE30529 (Kidney) · GSE14323 (Liver)
- GSE83717 (Lung RNA-seq) · GSE125362 (Skin Microarray)

## C. Consensus ML & WGCNA Discovery

### 5 Complementary Selection Architectures (5 Seeds)

1. LASSO (L1 Regularization)
2. SVM-RFE (Recursive Elimination)
3. Random Forest (Gini/Perm)
4. XGBoost (Gradient Boosting)
5. TOM Co-expression Clustering / WGCNA (Soft-thresholding)

### 9 Consensus Hub Genes (>= 3/5 Votes)

5 Unanimous Hubs (5/5 Votes):  
COL15A1 · COL1A1 · COL3A1 · SERPINE2 · SERPINF2  
+ 4 Extended Hubs: LAMC3 · LTBP2 · MDK · SVEP1

## D. Stage 3 Dual-Platform Validation & Locked 3-Tier Hub Hierarchy

### 3 Strict Validated Hubs

(p < 0.05 on Microarray & RNA-seq)

- COL15A1 (AUC: 0.843)  
Basement membrane organizer
- COL3A1 (AUC: 0.852)  
Early fibrillar repair collagen
- SERPINE2 (AUC: 0.786)  
Antiprotease/matrix protection

### 2 Direction-Concordant Hubs

(5/5 ML Votes; RNA-seq p >= 0.05)

- COL1A1 (AUC: 0.790)  
Canonical fibrillar collagen  
RNA-seq p = 0.101 (Concordant Up)
- SERPINF2 (AUC: 0.764)  
Alpha-2-antiplasmin inhibitor  
RNA-seq p = 0.788 (Concordant Down)

### 4 Stage-2 Candidates

(Failed Stage 3 Dual-Platform)

- LAMC3 (Discordant RNA-seq)
- LTBP2 (Discordant RNA-seq)
- SVEP1 (Discordant RNA-seq)
- MDK (RNA-seq p = 0.760)

Retained as pilot candidates

## E. Downstream Validation Modules

Mod 1: PPI Network (STRING v12)  
• High-confidence Collagen Triad (COL1A1-COL3A1-COL15A1)

Mod 2: Early-Detection Subgroup ROC (GSE84044)  
• S0 vs S1-S2 Subclinical AUC = 0.716 (5-Hub) / 0.719 (9-Hub)

Mod 3: Diagnostic Nomogram & DCA  
• Multivariable AUC = 0.9002 (5-Hub) / 0.9238 (9-Hub)

Mod 4: Immune & Stromal Infiltration  
• ACTA2 (Myofibroblast), POSTN (Matrix), PECAM1 (Endothelial)

Mod 5: HPA Baseline IHC & Biopsy RNA  
• Normal protein localization vs. patient transcript log2FC

Mod 6: GO / KEGG / DO Enrichment (Enrichr)  
• Extracellular matrix & collagen fibril organization